Crystallized Governance

MCIR, Jnana, and the Bounded Compilation of Constitutional Commitments

Crystallization can make selected governance commitments more explicit and testable.

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

Governed systems distribute their effective constitutions across prose, code, precedent, procedure, convention, identity, and material power. Crystallized Governance proposes that a bounded portion of this structure can be extracted into a canonical, inspectable, executable constitutional artifact using Meta-Compilation Intermediate Representation (MCIR) and the Jnana architecture.

The source claims that any governed system can be compressed into MCIR, that compression eliminates ambiguity, and that legitimacy becomes mechanical. The revised claim is narrower. Formalizable validity, authority, amendment, capability, effect, and fork rules may be compiled under a declared domain profile. Interpretation, value conflict, equitable discretion, tacit practice, contested identity, social power, and unanticipated cases may remain outside or unresolved. A responsible crystallization therefore carries a **semantic remainder** and an exact correspondence map back to source language and authority.

The paper defines a crystallization envelope binding source corpus, domain profile, extracted artifact, authority, translation map, remainder, evidence, and revision. It distinguishes the constitutional artifact from the process of Governance as Compilation and from Jnana II's transformation substrate. It adds contest, due process, privacy, exceptions, partial formalization, fork identity, GMB/GSM integration, and a full evaluation program.

Crystallization can make selected governance commitments more explicit and testable. It does not convert moral truth, legitimacy, interpretation, or social consent into a machine result, and no MCIR realization is established here.

Source, Evidence, and Standing

This is an AI-assisted integrated edition derived from one exact author manuscript. The custody and comparison manifest is

`work/paper-saturation-v1/evidence/`

`crystallized-governance/integration-custody-manifest.v0.1.json`,
SHA-256
`C7EC677DBB83BBD21DB7FBB11BB6D3AB64762A96EDE6AAB4FD9154D4BA3BF7A6`.
The source remains unchanged.

The crystallization thesis, MCIR and Jnana roles, transformation decomposition, compression concept, authority and fork structures, application profiles, and structural-clarity claim are author material. The bounded formalization, remainder, correspondence contract, corpus role distinctions, tests, counterevidence, and decisions added here are AI editorial or synthetic candidates until adopted. No external research was conducted.

1. Governance Is More Than Rules

A governed system includes written law and code, but also jurisdiction, authorship, precedent, interpretation, institutional capacity, exceptions, norms, incentives, enforcement, and affected-party rights. Some elements admit formal representation; others remain open-textured, contested, contextual, or inaccessible.

The thesis is therefore not that governance can be completely replaced by an executable artifact. It is that formalizable commitments should not remain ambient when they can be represented, tested, versioned, and disputed explicitly.

2. Crystallization Envelope

Define a governance crystallization as:

$$K = (N, p, C, A, M, U, E, R),$$

where N is the exact narrative and operational source corpus, p a domain and canonicalization profile, C the proposed constitutional artifact, A extraction and adoption authority, M a bidirectional correspondence map, U unresolved or intentionally uncompiled remainder, E evidence and tests, and R revision, appeal, and fork provisions.

The output is not only C. Without M and U, compression can erase the very ambiguity, exception, or conflict that governance must confront.

3. Partial Compilation

Crystallization may target:

- validity and eligibility predicates;
- state-transition and amendment procedures;

- role and capability grants;
- quorum, timing, and threshold rules;
- effect classes and approval boundaries;
- evidence and receipt obligations;
- exception and emergency procedures; and
- version, migration, and fork relations.

It may leave uncompiled:

- contested values and purposes;
- equitable discretion and proportionality;
- tacit institutional practice;
- ambiguous or conflicting source clauses;
- interpretation requiring human judgment;
- facts unavailable to the system; and
- novel cases outside the profile.

Uncompiled does not mean irrelevant. Remainder must have standing as remainder and a lawful route to interpretation or refusal.

4. MCIR and Canonical Representation

MCIR is proposed as a canonical Intermediate Representation for governed transformations. A profile must define syntax, semantics, serialization, equivalence, composition, validation, and evolution. Immutability, deterministic serialization, closure under composition, and executability are properties to demonstrate for a specific realization, not facts established by the paper.

Canonicalization produces stable identity only under its declared profile. Two source clauses may map to one artifact while losing distinctions; one clause may generate several interpretations; a source amendment may require a new profile. The correspondence map should preserve these cases rather than forcing false uniqueness.

5. Jnana Roles

Within the proposed architecture:

1. **MCIR** carries the constitutional representation.
2. **MCIR-Law** evaluates bounded admissibility questions.
3. **PEIR** realizes admitted procedures.
4. **Domain Effects** mediate declared external consequences.
5. **ReceiptLog** records identity, decisions, effects, and evidence.

Execution can still diverge from representation; effects can bypass a membrane; receipts can be incomplete; law can be unjust; and a canonical artifact can faithfully encode a bad constitution. The layers increase inspectability only where their correspondence and coverage are tested.

6. Authority Is Not Always One Lattice

The source models authority as a partially ordered set. This is useful where roles form stable dominance or delegation relations. Real systems may contain overlapping jurisdictions, separation of powers, context-dependent precedence, vetoes, mutual consent, temporary emergency authority, and unresolved conflicts.

Use an authority structure:

$$\mathcal{A} = (R, J, D, V, Q, X),$$

where R is roles, J jurisdictions, D delegation, V veto and consent relations, Q quorum and decision procedures, and X exceptions and conflicts. A lattice is one admissible profile, not a universal shape.

7. Admissibility and Refusal

A proposed amendment ΔC can be evaluated as:

$$G(C, \Delta C, L, A, E) \to \{admit, refuse, defer, escalate, not_covered\}.$$

The gate should bind law, authority, evidence, scope, effects, remainder, reason, expiry, and appeal. Invariant preservation is not self-proving; each invariant needs a test or proof class. Refusal preserves only the governed boundary if alternate effect paths remain unavailable.

8. Effect Membranes

An effect registry should declare classes rather than pretend to enumerate every future invocation. Each realized effect binds target, parameters, authority, outcome, partial failure, and recovery. Invocations outside the declared membrane may be refused by a conforming realization, but the architecture must separately test for bypasses through hosts, plugins, humans, credentials, and emergency paths.

No registry makes external action invalid in reality. It makes it unauthorized under the specified constitution.

9. Compression and Semantic Remainder

Compression can remove redundancy and expose equivalence. It can also erase context, minority interpretation, historical compromise, and useful ambiguity. Crystallization should report:

- source clauses represented exactly;
- normalized or merged clauses;
- inferred structure;

- conflicts and gaps;
- discretionary or human-judgment regions;
- excluded material and reason; and
- expected information loss.

The compression ratio is not a governance-quality metric. A longer artifact may be more honest because it preserves remainder.

10. Amendment, Fork, and Identity

Hard-fork-only, envelope-limited, fully revisable, and hybrid constitutions are candidate profiles. Fork identity depends on declared criteria and social recognition, not code alone.

An amendment record should bind predecessor, authority, procedure, evidence, effects, migration, compatibility, dissent, and appeal. A fork should state which identity, obligations, assets, records, and users carry forward. Competing identity claims may remain unresolved in GMB rather than being collapsed automatically.

11. Due Process and Contest

Executable governance can become faster and more consistent while also becoming rigid, inaccessible, or difficult to challenge. A constitutional artifact should expose:

- notice and intelligibility;
- affected-party participation;
- evidence and counterevidence;
- conflicts of interest;
- explanation and reason classes;
- appeal and human review;
- emergency and exception limits;
- accessibility and language needs; and
- amendment and exit.

Mechanical enforcement without contest can crystallize error and power as effectively as it crystallizes legitimate law.

12. Application Profiles

Cryptocurrency, DAO, corporate, regulatory, and AI-governance examples should be presented as profiles, not universal translations.

- A frozen store-of-value profile prioritizes limited mutation and reproducible validation but still depends on software, social, network, custody, and fork realities.
- A self-amending profile makes procedure explicit but does not guarantee legitimate participation or good amendments.

- A hybrid envelope freezes selected commitments while allowing bounded parameters to change; the boundary itself remains a constitutional issue.

Corporate and regulatory systems contain legal and human discretion that may not be executable without harmful loss. AI oversight additionally requires model, data, tool, authority, and effect boundaries.

13. Corpus Role Separation

Crystallized Governance concerns the constitutional artifact and its correspondence to source governance. **Governance as Compilation** concerns the process that translates proposal and law into decisions and effects. **Jnana II** supplies a candidate constitutional transformation substrate. **Jnana** supplies the original governed-genome architecture.

MCIR Topologies examines representational forms. **Meaning Crystals** supplies a multidimensional governed-memory concept rather than a constitutional artifact identity. **Governed Semantic Cartography** preserves semantic regions, disputes, and mappings. These relations are directional and do not merge the works.

The **Governed Meaning Base (GMB)** can retain sources, artifacts, maps, remainder, standing, disputes, interpretations, amendments, evidence, and forks. The universal **GSM** can type `COMPILES_FROM`, `REPRESENTS_CLAUSE`, `LEAVES_REMAINDER`, `AUTHORIZED_BY`, `CONTESTED_BY`, `AMENDS`, and `FORKS_FROM` without promoting translation into legitimacy.

14. Realization Funnel

$$\begin{aligned} & P \\ & \to S \\ & \to CM_1 \\ & \to CM_2 \\ & \to IC \\ & \to IR \\ & \to E \\ & \to R \end{aligned} .$$

- **Possibility:** enumerate full, partial, hybrid, frozen, revisable, and non-executable profiles.
- **Specification:** define source scope, profile, artifact, authority, mapping, remainder, effects, contest, and revision.
- **CM1:** retain admissible crystallizations, including incomplete, expensive, plural, or high-remainder forms.
- **CM2:** prefer configurations by fidelity, rights, inspectability, accessibility, safety, and governance cost.

- **IC:** select an Instance Constitution or Instance Configuration for one bounded governance domain.
- **IR:** produce an Instance Realization or Implementation Realization of schema, canonicalizer, gate, executor, membrane, receipt store, and interface.
- **Evidence:** test translation fidelity, ambiguity, rights, bypass, amendments, effects, appeals, and forks.
- **Revision:** amend source, profile, artifact, and mapping under explicit lineage.

This paper reaches Specification and proposes CM1/CM2. It does not select an IC, produce an IR, or establish an MCIR governance realization.

In this funnel, IR denotes the realization stage and remains distinct from Intermediate Representation in MCIR.

15. Evaluation Program

A pilot should select one bounded rule system with authoritative source text and real edge cases. Independent teams should extract the artifact and remainder; domain experts and affected users should review correspondence; adversarial cases should test conflicts, gaps, bypass, exceptions, partial effects, appeal, amendment, and fork identity.

Measures should include clause coverage, inferred versus explicit structure, unresolved remainder, inter-extractor agreement, false admission and refusal, effect coverage, appeal reversals, accessibility, maintenance cost, semantic drift, and user understanding. Compare against executable policy, conventional code, and narrative governance baselines.

Constitutional Artifact and Correspondence Profile

Across the completed P1 corpus, Crystallized Governance is the **constitutional artifact and source-correspondence profile for bounded partial formalization of governance**. It asks which commitments can be made explicit, canonical, testable, and revisable while carrying disputed interpretation, uncompiled remainder, due process, and amendment authority beside the artifact. It is not the governance-compilation process, the execution substrate, or a claim that legitimacy becomes mechanical.

- **Adjacent work:** *Governance as Compilation*
  Supplies: The staged translation process from source meaning and proposal toward decision, plan, effect, and evidence.
  Crystallized-Governance boundary: Crystallized Governance specifies the constitutional artifact, its sources, map, remainder, and revision identity.
- **Adjacent work:** *Jnana II: Constitutional Transformation Substrate*

Supplies: A candidate substrate for transformation admission, capabilities, effects, and receipts.
Crystallized-Governance boundary: A substrate may realize a crystal profile but cannot supply its legitimate source authority.

- **Adjacent work:** *MCIR Topologies*
  Supplies: Configuration grammar for canonical representational structures.
  Crystallized-Governance boundary: Topology determines representation choices, not fidelity, adoption, or justice.
- **Adjacent work:** *Governed Semantic Cartography*
  Supplies: Mapping of semantic regions, disputes, boundaries, and plural interpretations.
  Crystallized-Governance boundary: Cartography can preserve the terrain from which a bounded constitutional artifact is selected.
- **Adjacent work:** *Meaning Crystals*
  Supplies: A wider model of stable, reopenable, multidimensional semantic memory.
  Crystallized-Governance boundary: A governance crystal is one specialized artifact class with rights and effect consequences.

The constitutional crystal should be treated as a compound object:

$$K = (sources, profile, artifact, correspondence, remainder, \\ authority, rights, interpretation, effects, evidence, \\ contest, amendment, forks) \quad .$$

`correspondence` must be many-to-many: one clause can support several interpretations, multiple sources can jointly support one rule, and a canonical rule can compress distinctions that remain material.

`remainder` is neither noise nor automatic refusal; it can be disputed, intentionally discretionary, protected, unavailable, or outside the profile.

The profile should distinguish representational validity, source fidelity, authorized adoption, procedural application, effect conformance, and continuing legitimacy. Success at any earlier layer does not entail the later layers. A perfectly faithful artifact can encode an unjust constitution; a just source can be mistranslated; an adopted artifact can be bypassed in effect.

The universal GSM can type
`CRYSTALLIZES_GOVERNANCE_SOURCE`,
`CORRESPONDS_TO_SOURCE_CLAUSE`, `COMPRESSES_SOURCE_DISTINCTION`,
`PRESERVES_SEMANTIC_REMAINDER`, `ADOPTED_BY_AUTHORITY`, `CONTESTED_BY_AFFECTED_PARTY`,
`INTERPRETED_UNDER_PROFILE`, `AMENDS_CONSTITUTIONAL_ARTIFACT`, `FORKS_GOVERNANCE_IDENTITY`,
and

REMAINS_DISTINCT_FROM_EXECUTION_SUBSTRATE. These relations preserve the paper's specific role.

This integration establishes four editorial commitments: retain Crystallized Governance as the constitutional artifact-and-correspondence profile; adopt the expanded compound object and many-to-many mapping; separate validity, fidelity, adoption, application, effect, and legitimacy claims; and preserve remainder, contest, rights, amendment, and fork identity as first-class constitutional structure.

From Restored Constitutional Artifact to Admissible Effect

The fixed-seed comparison with Cold Boot Continuity, the Cognitive ABI, Admissibility Engineering, and the international-trade UGK profile separates constitutional restoration from operational effect:

$$G = (sources, artifact, correspondence, remainder, \\ current_adoption, interpretation, capability_contract,. \\ pathway_constraints, effect, evidence, contest)$$

Cold boot can restore the exact artifact, sources, correspondence map, amendment lineage, contest state, and remainder. It cannot infer that an earlier adoption, delegation, or interpretation remains live. Current jurisdiction, revocation, affected parties, and amendment posture must be revalidated.

The Cognitive ABI can expose operations governed by the artifact. Interface conformance is not constitutional fidelity or authority. Admissibility Engineering then asks whether the artifact's consequential obligations bind every discovered realization pathway and whether legitimate variation, refusal, appeal, and remainder survive implementation.

The international-trade material is a possible domain stress profile because it contains multi-party mandates, handoffs, amendments, jurisdictional conflict, and legal-effect boundaries. Its source-title and proposed-title mismatch remains on identity review; no domain relation in this pass merges titles, creates publication identity, or establishes legal standing.

The universal GSM can type
RESTORES_CONSTITUTIONAL_ARTIFACT,
REVALIDATES_CURRENT_ADOPTION, EXPOSES_GOVERNED_CAPABILITY,
BINDS_CONSTITUTIONAL_PATHWAY, PRESERVES_CONTEST_ON_EFFECT, PROFILES_TRADE_HANDOFF,
and
HOLDS_DOMAIN_IDENTITY_FOR_AUTHOR_REVIEW.

This randomized review establishes four editorial commitments: adopt the restoration-to-effect chain; require current adoption and interpretation validation; connect ABI operations to pathway-constraint evidence; and retain the trade profile only as an identity-held domain stress candidate.

The Constitutional Artifact as a Versioned Consumer Contract

Five downstream uses clarify what a crystallized governance artifact must supply. Admissibility Engineering consumes source correspondence, remainder, adoption, contest, and amendment obligations while testing realized pathways. Cold boot may restore the artifact but must revalidate current grants and environment. A Cognitive ABI consumes selected constraints in capability qualification and invocation. Governance as Compilation produces or transforms the artifact but is not identical to it. The international-trade profile applies mandates, documentary correspondence, jurisdiction, contest, and amendment to a bounded domain.

The constitutional artifact and its consumer binding should be:

$$\begin{aligned} K = (&artifact_version, sources, translation_map,\\ &adopted_content, remainder, exceptions, jurisdiction, ,\\ &authority, contest, expiry, amendment, fork, evidence) \end{aligned}$$

$$\begin{aligned} U = (&consumer, purpose, pinned_version, applied_clauses,\\ &interpretation, local_authority, omissions,\\ &dependencies, revalidation, invalidation,\\ &effect_ceiling) \end{aligned} .$$

The artifact is not executable authority. Each consumer must bind the exact version, purpose, clauses used, interpretation, local authority, omitted material, and conditions requiring revalidation. Two consumers may lawfully interpret or apply different subsets under different jurisdictions without creating one ambient constitution. A restored or hash-identical artifact can be stale because authority, law, dependencies, evidence, or environment changed.

Amendment and fork decisions should identify affected consumers and whether they pin the predecessor, follow an approved head, require re-adoption, or become historical only. Domain profiles may specialize the artifact but cannot inherit legal, professional, operational, or effect authority from structural conformance.

The universal GSM can type
`CRYSTALLIZES_CONSTITUTIONAL_ARTIFACT`,
`BINDS_CONSUMER_TO_VERSION`, `APPLIES_CLAUSE_FOR_PURPOSE`, `OMITS_WITH_DECLARED_REMAINDER`,
`REVALIDATES_RESTORED_ARTIFACT`, `INVALIDATES_CONSUMER_BINDING`,
`FORKS_GOVERNANCE_ARTIFACT`, and
`CONFORMS_WITHOUT_EFFECT_AUTHORITY`.

This reverse-dependency review establishes four editorial commitments: adopt the artifact and consumer-binding pair; require exact purpose, clause, interpretation, and omission records; revalidate restored artifacts against current authority and environment; and propagate amendment or fork effects to consumers without ambient authority transfer.

16. Counterevidence and Failure Modes

The proposal should be narrowed or rejected where: (1) any governed system is claimed fully representable; (2) narrative governance is treated as mere ambiguity; (3) compression erases context or minority meaning; (4) remainder has no standing; (5) translation lacks exact source custody; (6) canonicalization is treated as truth; (7) MCIR properties are asserted without implementation; (8) execution is assumed to preserve representation; (9) authority is forced into one lattice; (10) jurisdictional conflict is erased; (11) invariant preservation is untested; (12) admissibility is confused with legitimacy; (13) effect bypasses remain; (14) an unauthorized action is called impossible; (15) receipts are treated as justification; (16) procedural consistency crystallizes injustice; (17) human discretion is eliminated where necessary; (18) exception paths become ambient; (19) due process is absent; (20) appeals cannot alter outcomes; (21) accessibility burdens exclude affected people; (22) amendment authority captures the constitution; (23) fork identity is reduced to hashes; (24) cryptocurrency profiles ignore social and network reality; (25) corporate or regulatory law is oversimplified; (26) GMB inclusion becomes standing; (27) corpus roles are merged; (28) a specification is described as executable; (29) extraction teams cannot agree; (30) semantic drift exceeds the baseline; (31) governance cost outweighs benefit; or (32) external research shows existing approaches already cover the claimed novelty.

Formalize What Can Be Formalized and Preserve What Cannot

Public orientation

Crystallized Governance proposes that formalizable constitutional commitments should become explicit, versioned, testable artifacts rather than remain ambient in prose and implementation. A responsible crystal also carries exact sources, a correspondence map, authority, unresolved remainder, contest, amendment, and fork paths. It is not the whole constitution and does not turn legitimacy into a machine result.

Adversarial tests

1. **Compression-erasure attack.** Ambiguity, exception, minority meaning, historical context, or contested interpretation disappears from the artifact.
2. **Correspondence-fiction attack.** A source-to-artifact map exists syntactically but does not show entailment, authorized interpretation, omissions, or semantic loss.
3. **Artifact-authority attack.** Possession, hash identity, canonical form, or successful validation is treated as present authority to govern a consumer.
4. **Bad-law compilation attack.** A coherent and faithfully encoded constitution is unjust, unauthorized, obsolete, or procedurally illegitimate.

5. **Discretion-elimination attack.** Equity, proportionality, tacit practice, novel cases, and human judgment are forced into rules that cannot responsibly decide them.
6. **Consumer-drift attack.** A consumer applies the wrong version, purpose, clause subset, interpretation, jurisdiction, or expired authority.
7. **Exception-capture attack.** Emergency or exception paths become ambient bypasses without expiry, review, appeal, or evidence.
8. **Fork-hash attack.** Competing governance communities or successor identities are reduced to artifact hashes while adoption, consent, jurisdiction, and social continuity disappear.

Claim and evidence ceiling

The paper proposes bounded partial compilation through a versioned constitutional artifact and consumer-binding contract. It does not establish complete representability, moral truth, legitimacy, legal validity, consent, fair enforcement, executability, MCIR conformance, or effect authority. Uncompiled remainder is a first-class governed result, not failed data awaiting forced normalization.

Legal, political, institutional, organizational, cryptographic, usability, and social claims require current external research, authorized domain interpretation, affected-party review, exact realizations, and evaluated contest paths. No external research was added in this pass.

Reader-facing consequence

The public promise is not government by code. It is that whenever governance is made executable, people can see the exact source, translation, authority, omitted remainder, consumer purpose, challenge path, and conditions that require re-adoption or refusal.

18. Conclusion

Crystallized Governance proposes a valuable shift: make formalizable constitutional commitments explicit, canonical, testable, and versioned. The artifact can clarify validity, authority, amendment, capability, effect, and fork rules that would otherwise remain distributed and implicit.

The artifact is not the whole constitution. Legitimate governance also contains interpretation, contest, history, discretion, rights, and social authority. A faithful crystal therefore includes its own boundary: exact sources, translation map, unresolved remainder, evidence, and revision path.

Governance does not become legitimate merely because it is executable. It becomes more inspectable where the executable structure accurately carries authorized commitments and remains open to evidence, contest, and change.

Reference and Verification Note

No external source was searched in this pass. Further evidentiary strengthening requires authorized primary-source review of constitutional and administrative law, legal formalization, executable policy and smart contracts, DAOs and cryptocurrencies, corporate governance, regulatory practice, authority and delegation models, formal methods, canonicalization, due process, accessibility, and existing policy-language systems.

Realization Scope and Evidence Boundary

This successor makes the paper's stage boundary explicit. It narrows claims rather than manufacturing missing evidence: a stage named as context is not thereby claimed as implemented, validated, or complete.

- **Possibility Space.** The present paper does not claim a complete possibility space treatment. That stage remains successor work and cannot be inferred from the architecture described here.

Corpus-Wide Reference Positioning

This paper argues that governed systems distribute their effective constitutions across prose, code, precedent, procedure, convention, identity, and material power. Its nearest external comparison fields are governance, formal, provenance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. A complete provenance graph can faithfully record a false or unauthorized process; provenance is evidence about lineage, not automatic truth or legitimacy. These

limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action.* Cambridge University Press, 1990. DOI: 10.1017/CBO9780511807763. https://doi.org/10.1017/CBO9780511807763 **Relationship:** Predecessor institutional account of rule-governed collective action and polycentric governance.

[**E2**] Wil Waluchow and Dimitrios Kyritsis. "Constitutionalism." *Stanford Encyclopedia of Philosophy*, substantive revision 2023. https://plato.stanford.edu/entries/constitutionalism/ **Relationship:** Adjacent legal-philosophical account of limited authority; included to expose differences between political and computational constitutionalism.

[**E3**] Lawrence Lessig. *Code: Version 2.0.* Basic Books, 2006. https://codev2.cc/ **Relationship:** Predecessor argument that software architecture regulates behavior; the paper's contribution may be to make such regulation inspectable and revisable.

[**E4**] OECD. *Recommendation of the Council on Artificial Intelligence.* OECD/LEGAL/0449, adopted 2019 and updated 2024. legalinstruments.oecd.org source **Relationship:** Adjacent policy standard for accountability, traceability, transparency, safety, and human-centered values.

[**E5**] George C. Necula. "Proof-Carrying Code." *POPL '97*, 1997, pp. 106-119. DOI: 10.1145/263699.263712. https://doi.org/10.1145/263699.263712 **Relationship:** Predecessor for binding machine-checkable evidence to executable artifacts; the present paper may demand broader semantic or constitutional obligations.

[**E6**] Xavier Leroy. "Formal Verification of a Realistic Compiler." *Communications of the ACM* 52(7), 2009, pp. 107-115. DOI: 10.1145/1538788.1538814.

https://doi.org/10.1145/1538788.1538814 **Relationship:** Predecessor verified-compilation result used to distinguish proven translation preservation from broader governance claims.

[**E7**] Patrick Cousot and Radhia Cousot. "Abstract Interpretation: A Unified Lattice Model for Static Analysis of Programs by Construction or Approximation of Fixpoints." *POPL '77*, 1977. DOI: 10.1145/512950.512973. https://doi.org/10.1145/512950.512973 **Relationship:** Predecessor formal framework for sound approximation and fixpoint reasoning.

Ableman corpus

[**A1**] Adam Ableman. *Ableman Constitutional Governance*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-79353AF163A4B0F1`;
source SHA-256
`FB0D9B2D3D54168FFE70B24308D1B2766D6D663ED24502D475E975939BE97C8B`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Constitutional Theory*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D2D97C4042FD281F`;
source SHA-256
`FACC256EDDE5A7D7EA18133CFC9C060CFB8C3811B0221E1969E62347F2BB69CF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Governance Engineering*. Ableman Corpus current edition; work identity `PAPER.PUB.18749870`;
source SHA-256
`372169F315D192733856B1F4A3F1038BBC73F57CF3ED9ACABAE72B965A69C8D0`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Universal Governance Kernel (UGK)*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-1F9BFB2BF8370226`;
source SHA-256
`AA5106EA71E5DF120E141435C280DFC532715CA5DE0C75B8862BB7784ECCFACD`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Legitimacy Without Self-Ratification*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-EF2382F9397335AC`;
source SHA-256
`0186A39C808F8FDDB5C8F42384DF1A26678928434BD50A0EAE8D4947390F68A2`.

Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Ableman Corpus current edition; work identity `PAPER.PUB.21095303`;
source SHA-256
`EA129ECBA95F7B07CAA7B33A54F91F52913C94A14DA227EB9A06CFD083E4E5BF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Governed Semantic Compilation.* Ableman Corpus current edition; work identity `PAPER.PUB.18453681`;
source SHA-256
`5FF6C636550B04DB0E2F6F94C379BB75AA0BD8781E19B047D7D3BB6D49071AC2`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *A Theory of Agentic Observation.* Zenodo, 2026. doi:10.5281/zenodo.20437813

Adam Ableman *CARA: Constrained Artificial Reasoning Architecture.* Zenodo, 2025. doi:10.5281/zenodo.18099715

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Adam Ableman *Governed Semantic Compilation: Lawful Continuation, Refusal Semantics, and Epistemic Promotion in Large Language Models.* Zenodo, 2026. doi:10.5281/zenodo.18453681

Adam Ableman *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Zenodo, 2026. doi:10.5281/zenodo.21095303

Adam Ableman *Semantic Coordinate Identity Tokenization (SCIT).* Zenodo, 2026. doi:10.5281/zenodo.20532070

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.23847625